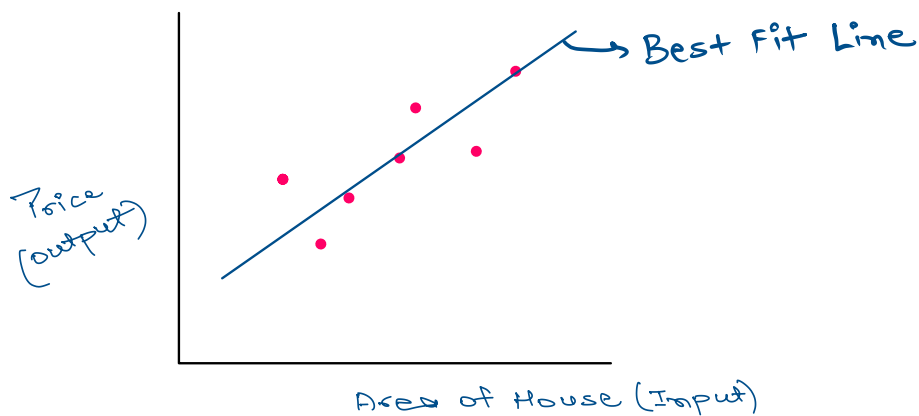


many lines passing through the data points, then calculate "Cost Function for each line" which is nothing but the total error made by each line. Then it compares the Cost Functions $\rightarrow$

$$CF(\text{line1}) \text{ vs } CF(\text{line2}) \text{ vs } CF(\text{line3}) \text{ vs } \dots$$

Now, whichever line has least cost function will be our "Best Fit Line".



This line can be represented as an equation which is in this form $\rightarrow$

$$y = mx + c$$

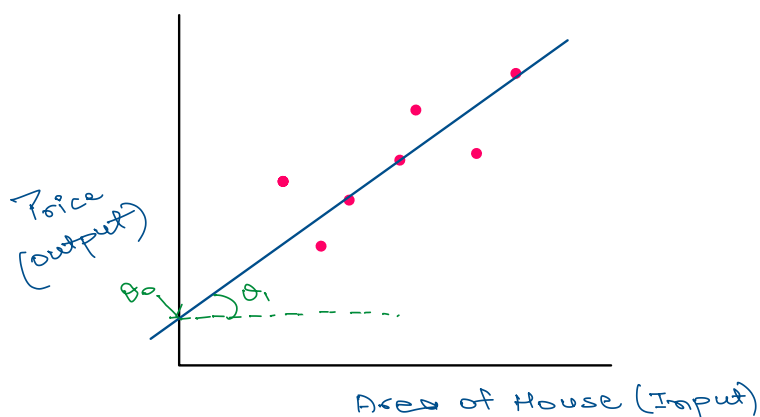
that we write as:

$$\hat{y} = \theta_0 + \theta_1 x$$

and in final form it will look like $\rightarrow$

$$\text{Predicted Price } (\hat{y}) = 1.8 + 0.9x \quad (\text{suppose})$$

where the value of θ_0 and θ_1 will be taken from the best fit line and x is the future input.



In this process, the algorithm has to try 100s of combination of θ_0 & θ_1 to find the ones that minimize the cost function.

Multiple Linear Regression $\rightarrow$

x_1 No of bedrooms	x_2 Age	x_3 No of floors	x_4 Area of House	output col $\downarrow$ Price

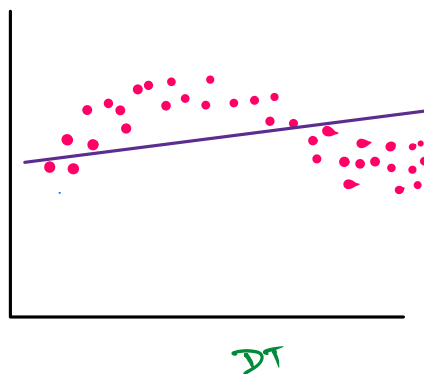
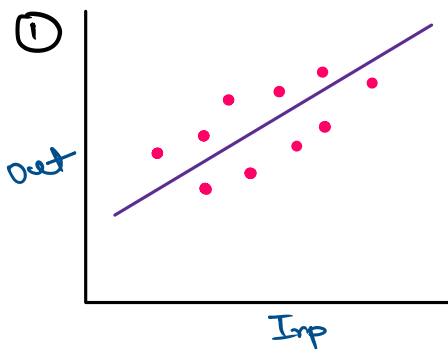
$$\hat{y} = w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4 + c$$

$$= 1.1 \times \text{No of Bedrooms} + (-1.26) \times \text{Age of House} + 1.8 \times \text{No of Floors} +$$

$$2.1 \times \text{Area of House} + \underline{1.19}$$

$\uparrow$
Base Price

Assumptions $\rightarrow$

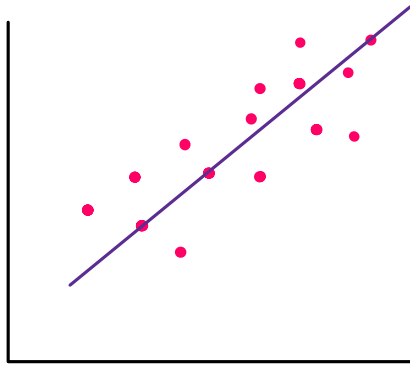


② $e_1 + e_2 + e_3 + e_4 + \dots$

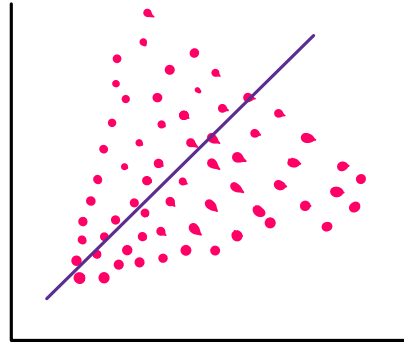
↓
zero or close zero

Then you can use Linear Regression on this data.

③



Homoscedasticity ✓
✓

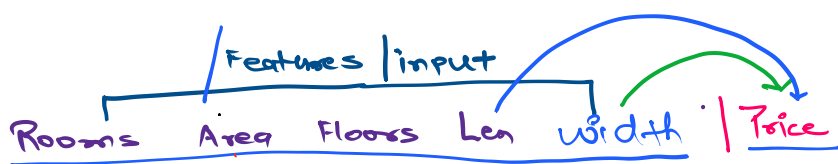


Heteroscedasticity ✗

④ 'Multicollinearity' should not be present in the data.

↓
When input cols are highly correlated to each other that is called as multicollinearity.

redundant



VIF (Variance Inflation Factor) helps in

identifying and removing multicollinearity.

VIF score > 5

→ That col is having >80% correlation with other columns.